STAT 405 Project Proposal

Team Member:

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Github website:

https://github.com/HaoyuYou/Bayesian-Statistics-Project.git

Dataset

main: Bike Sharing Demand

The dataset used in this project is the Bike Sharing Dataset from the UCI Machine Learning Repository. It contains information about daily bike rental usage in a bike sharing system from 2011 to 2012.

We use the daily dataset (`day.csv`), which includes 731 observations, where each observation represents one day. The dataset records the total number of bike rentals (`cnt`) along with several explanatory variables related to time and weather conditions.

Key variables include temperature (`temp`), humidity (`hum`), wind speed (`windspeed`), season (`season`), weather situation (`weathersit`), and whether the day is a working day (`workingday`). These variables help explain patterns in bike rental demand.

```r
bike <- read.csv("data/day.csv")
cat_vars <- c("season", "yr", "mnth", "holiday", "weekday", "workingday", "weathersit")
bike[cat_vars] <- lapply(bike[cat_vars], as.factor)
head(bike)
```

```
##   instant     dteday season yr mnth holiday weekday workingday weathersit
## 1       1 2011-01-01      1  0    1       0       6          0          2
## 2       2 2011-01-02      1  0    1       0       0          0          2
## 3       3 2011-01-03      1  0    1       0       1          1          1
## 4       4 2011-01-04      1  0    1       0       2          1          1
## 5       5 2011-01-05      1  0    1       0       3          1          1
## 6       6 2011-01-06      1  0    1       0       4          1          1
##       temp    atemp      hum windspeed casual registered  cnt
## 1 0.344167 0.363625 0.805833 0.1604460    331        654  985
## 2 0.363478 0.353739 0.696087 0.2485390    131        670  801
## 3 0.196364 0.189405 0.437273 0.2483090    120       1229 1349
## 4 0.200000 0.212122 0.590435 0.1602960    108       1454 1562
## 5 0.226957 0.229270 0.436957 0.1869000     82       1518 1600
## 6 0.204348 0.233209 0.518261 0.0895652     88       1518 1606
```

URL:

https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset

backup: BMW Global Automotive Sale

```
bmw <- read.csv("data/bmw_global_sales_2018_2025.csv")
head(bmw)
```

```
##   Year Month Region    Model Units_Sold Avg_Price_EUR Revenue_EUR BEV_Share
## 1 2018     1 Europe 3 Series       7822         47482   371404204     0.011
## 2 2018     1 Europe 5 Series      10280         61685   634121800     0.019
## 3 2018     1 Europe       X3       3105         58433   181434465     0.022
## 4 2018     1 Europe       X5       7420         67955   504226100     0.021
## 5 2018     1 Europe       X7       8474         92300   782150200     0.035
## 6 2018     1 Europe       i4      10607         64082   679717774     0.035
##   Premium_Share GDP_Growth Fuel_Price_Index
## 1         19.12        3.5                1
## 2         19.12        3.5                1
## 3         19.12        3.5                1
## 4         19.12        3.5                1
## 5         19.12        3.5                1
## 6         19.12        3.5                1
```

URL:

https://www.kaggle.com/datasets/dmahajanbe23/bmw-global-automotive-sales

Scientific Question:

How do weather conditions and seasonal factors influence daily bike rental demand? Does a more flexible Bayesian count model provide a better fit and more realistic uncertainty quantification than a simpler baseline model?

More specifically, we aim to model how variables such as temperature, humidity, wind speed, season, and working day status affect the total number of bike rentals (`cnt`).

Bike sharing demand is influenced by both environmental conditions and human behavior patterns. Understanding these relationships can help improve demand prediction and support better planning for urban transportation systems.

Using Bayesian modelling allows us to quantify uncertainty in parameter estimates and make probabilistic interpretations of the effects of different predictors on bike rental demand.

Methodology:

We plan to build Bayesian models to explain and predict the number of daily bike rentals. In particular, we will consider two models: a simple baseline model and a more flexible regression model incorporating environmental predictors.

Before fitting the Bayesian models, we will perform exploratory data analysis (EDA) to understand the distributions of key variables and their relationships with bike rental demand. This includes visualizing the distribution of the response variable (`cnt`) and examining how it varies with weather conditions and seasonal factors.

Model 1: Naive Model

As a baseline, we first consider a simple Bayesian Poisson regression model for daily bike-sharing demand. Let Y_i be the total rental count (`cnt`) on day i. Since the response variable is a nonnegative count, a Poisson model provides a natural starting point for modeling the distribution of daily demand. We assume

$$Y_i|\lambda_i \sim \text{Poisson}(\lambda_i) \qquad \log(\lambda_i) = \beta_0 + \beta_1 temp_i + \beta_2 hum_i + \beta_3 windspeed_i$$

In this model, λ_i represents the expected number of bike rentals on day i, and its logarithm is modeled as a linear function of three continuous weather-related predictors: temperature, humidity, and windspeed. The log link is used to ensure that the Poisson mean λ_i is always positive.

Model 2: Complex Model

As a more flexible model, we then consider a Bayesian hierarchical Poisson regression model with month-specific intercepts. The purpose of this hierarchical structure is to capture systematic variation in baseline demand across months while allowing partial pooling across groups. This allows the model to borrow strength across months and produce more stable estimates.

For the other calendar-related factors, we treat them as fixed effects because of their small number of groups, even if some of them are relatively imbalanced. Because season is largely determined by month, including both month effects and season effects would introduce substantial redundancy and may lead to aliasing issues. Therefore, once month is used as the main grouping factor in the hierarchical model, season is omitted from the linear predictor. Therefore, for this hierarchical model, we assume

$$Y_i \mid \lambda_i \sim \text{Poisson}(\lambda_i),$$
$$\log(\lambda_i) = \alpha_{m[i]} + \beta_1 \, \text{temp}_i + \beta_2 \, \text{hum}_i + \beta_3 \, \text{windspeed}_i + \beta_4 \, \text{holiday}_i + \beta_5 \, \text{workingday}_i + \beta_6 \, \text{weathersit}_i + \beta_7 \, \text{yr}_i,$$

where $m[i] \in \{1, \ldots, 12\}$ indicates the month associated with observation i. The month-specific intercepts are modeled hierarchically as

$$\alpha_m \sim N(\mu_\alpha, \sigma_\alpha^2), \qquad m = 1, \ldots, 12.$$

We assign weakly informative priors to the regression coefficients and hyperparameters, for example,

$$\beta_j \sim N(0, 10^2), \qquad j = 1, \ldots, 7,$$
$$\mu_\alpha \sim N(0, 10^2), \qquad \sigma_\alpha \sim \text{HalfNormal}(0, 5).$$

This hierarchical Bayesian framework allows us to model both the overall effects of weather and calendar-related predictors and the month-to-month heterogeneity in baseline demand.

Inference

The proposed models will be implemented in Stan. Posterior inference will primarily be performed using Markov chain Monte Carlo (MCMC) sampling.

To satisfy the course requirement of using two posterior inference methods, we also plan to compare MCMC-based inference with a simpler approximation-based approach, such as MAP/Laplace approximation.

From the posterior output, we will estimate:

- posterior distributions of model parameters
- uncertainty in expected and predicted bike rental demand
- month-specific baseline demand effects in the hierarchical model
- the effects of weather and calendar-related predictors on bike-sharing demand

Finally, we will compare the naive and hierarchical models using posterior predictive checks and held-out predictive performance. We will also compare the two inference approaches in terms of posterior summaries, uncertainty quantification, and predictive behavior.

Contribution plan:

This is a team project and both team members will contribute to all stages of the project.

Ruohan Sun

- Data preprocessing and exploratory data analysis (EDA)
- Implementation of the Bayesian models
- Posterior inference and model diagnostics
- Visualization of results

Haoyu You

- Literature review and background research
- Model formulation and prior specification
- Model comparison and interpretation of results
- Writing and editing the final report